

Floor-Free Semidefinite Directional Transport Rank Creation, Exact-Gram Minimax, and a Five-Scale Audit

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Abstract

The exact-Gram gauge audit of RH-121 and the combined directional chains of RH-125 inserted positive conditioning floors into both the recent Gramian and the tail pencil. We remove those floors and extend the exact-Gram minimax theorem from positive definite to positive semidefinite tails. If $\alpha_1 \leq \dots \leq \alpha_m$ and $\beta_1 \leq \dots \leq \beta_m$ are the two normalized tail spectra, then the least multiplicative tail factor remains

$$b_* = \max_i \frac{\beta_i}{\alpha_i},$$

with the decisive convention that $\beta_i/0 = +\infty$ when $\beta_i > 0$. Consequently a finite exact-Gram transfer exists only if tail rank cannot increase. A positive floor erases this obstruction and may replace $+\infty$ by an arbitrarily large finite number.

We then rebuild the five-scale packet through one common assembly, form the reduced 4×4 Gramians at 90 decimal digits, and add no positive Gram or tail floor. All 120 recent actions retain exact discrete rank four and all four directions are separated at a conservative fp64 conditioning scale. Moreover, 118 of 120 single-scale directional candidates remain positive. Cross-scale transport is different: only 67 of 96 adjacent transfers are positive, 24 have infinite optimal factor because a new tail direction appears, five further finite transfers have $\gamma \geq 1$, and none of the 24 four-edge chains survives. Thus the encouraging regularized chain count was not floor-stable. The negative result does not close the directional route: it identifies the missing mechanism precisely as an additive forcing term, rather than loss of local four-directional support.

1 Why the floor question is structural

Let A be a four-column recent action, $G = A^*A$, and let $D \succeq 0$ be a tail Gram upper bound. The relative Rayleigh constant is the least γ for which

$$D \preceq \gamma^2 G.$$

When $G \succ 0$, this is the square root of the largest eigenvalue of $G^{-1/2} D G^{-1/2}$. The four-volume certificate used in RH-114 and RH-125 is

$$B(G, D; L, C) = \frac{(1 - \gamma)_+^4 \sqrt{\det G}}{L^4 C}, \quad (x)_+ = \max\{x, 0\}.$$

Here L is a leading singular-value upper bound and C is the capacity normalization. This formula is exact for the finite packet, but its value can be changed qualitatively by replacing G, D with $G + \varepsilon_G I$ and $D + \varepsilon_D I$.

The issue is not merely that a small eigenvalue is difficult to compute. The discarded memory tail is genuinely semidefinite. Before the memory depth is exceeded it is exactly zero; later it can acquire rank one direction at a time. A positive tail floor changes each zero eigenvalue into a positive one and therefore changes the set of admissible Loewner comparisons. In particular it can make a multiplicative recurrence appear finite at the exact instant at which the true semidefinite recurrence needs an additive source.

2 Semidefinite exact-Gram minimax

We first close the finite-dimensional theorem without assuming that either tail is positive definite. Let $G, G' \succ 0$ and $D, D' \succeq 0$. Write

$$A = G^{-1/2} D G^{-1/2}, \quad B = G'^{-1/2} D' G'^{-1/2},$$

and denote their eigenvalues in nondecreasing order by α_i and β_i .

Theorem 1 (Semidefinite exact-Gram minimax). *Among all invertible gauges S satisfying $S^* G S = G'$, the least $b \in [0, +\infty]$ for which*

$$D' \preceq b S^* D S$$

is

$$b_* = \max_{1 \leq i \leq m} \beta_i / \alpha_i,$$

where $0/0$ contributes no constraint and $x/0 = +\infty$ for $x > 0$. The infimum is attained by matching the ordered eigenframes. In particular, $b_ < \infty$ if and only if*

$$\text{rank } D' \leq \text{rank } D.$$

Proof. Every exact-Gram gauge has the form $S = G^{-1/2} U G'^{1/2}$ with U unitary. Conjugating the desired inequality by $G'^{-1/2}$ gives

$$B \preceq b U^* A U.$$

Loewner monotonicity of ordered eigenvalues implies $\beta_i \leq b \alpha_i$ for every i , including the stated infinite obstruction when $\alpha_i = 0 < \beta_i$. Hence every feasible b is at least b_* . Choose U to map the ordered eigenbasis of B to that of A . The comparison becomes the diagonal family $\beta_i \leq b_* \alpha_i$ and is therefore attained. Since the zero eigenvalues occur first, finiteness is equivalent to the target having at least as many zero eigenvalues as the source, or equivalently $\text{rank } D' \leq \text{rank } D$. $\square$

The theorem is the semidefinite completion of the RH-121 positive-definite formula. It also shows why replacing a zero source eigenvalue by a tiny number is not an innocent numerical convention.

Corollary 2 (Rank-creation obstruction). *If an adjacent scale creates a tail direction, $\text{rank } D' > \text{rank } D$, then no exact-Gram gauge and no finite multiplicative factor can transfer D to D' . Any valid recurrence must contain an additive positive term whose range covers the newly created direction.*

Proposition 3 (Sharp floor instability). *For every $M > 0$ there are scalar or diagonal semidefinite pencils for which the floor-free optimal factor is infinite while the ε -floored factor is finite and exceeds M . More precisely, for $A = \text{diag}(0, a_2, \dots, a_m)$ and $B = \text{diag}(\beta, b_2, \dots, b_m)$ with $\beta > 0$, the first ordered ratio after replacing A by $A + \varepsilon I$ is β/ε .*

Proof. The floor-free statement follows from Theorem 1. After the floor, the formerly impossible comparison is the scalar inequality $\beta \leq b\varepsilon$, whose sharp factor is β/ε . Choosing $\varepsilon < \beta/M$ gives the claim. $\square$

Thus convergence of the floored matrices in norm does not imply convergence of the optimal transport factor. Rank is a discontinuous datum at the boundary of the positive cone, as expected from standard perturbation theory [Kato, 1995, Horn and Johnson, 1991].

3 Common-assembly audit

The numerical audit uses exactly the five frozen scales $\sigma \in \{0.16, 0.08, 0.04, 0.02, 0.01\}$, both left and right channels, three packet thresholds, and four phase locations. This gives 120 states, 96 phase-matched adjacent-scale pairs, and 24 four-edge chains. In contrast to RH-121, the recent and discarded histories are assembled directly as disjoint positive sums; the tail is not obtained by subtracting two nearly equal full Gramians. The same action, frame, tail block, leading bound, and capacity bound are then used for every downstream quantity.

The large frozen model is propagated in fp64, while each reduced Gramian is formed from its stored action entries and diagonalized at 90 decimal digits. This targets the actual cancellation point without pretending that the upstream discretization has 90-digit physical accuracy. Separately, a conservative fp64 support diagnostic declares a singular direction visible only above

$$r_A = 128 \epsilon_{64} \max(m, 4) \|A\|_2.$$

The two ranks answer different questions: exact rank of the frozen discrete data and rank stable at the declared arithmetic scale [Higham, 2002]. No positive value is added to G or D . The analytic geometric-memory bound supplies δ in the positive inequality $R^*R \preceq \delta B$; the audit records the minimum high-precision slack.

4 Results: local support survives, multiplicative chains do not

All 120 recent actions have exact discrete rank four, and all 120 retain rank four under the declared fp64 support radius. The largest action condition number is $10^{9.9844}$; the median logarithmic condition number is about 3.25. These values explain why Gram eigenvalues seen

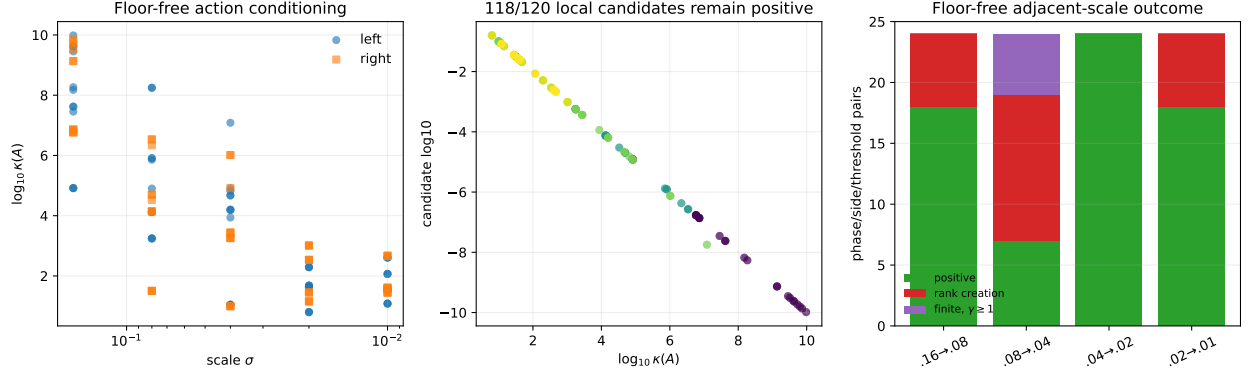


Figure 1: Left: conditioning of all 120 floor-free actions. Center: local candidate size versus conditioning. Right: the 96 adjacent-scale outcomes; rank creation denotes an infinite exact-Gram tail factor.

in ordinary precision can be much smaller—conditioning is squared by A^*A —but they do not indicate actual loss of the fourth action direction.

The local Rayleigh calculation is also mostly stable under floor removal. Exactly 118 of 120 states have $\gamma < 1$ and therefore a positive directional candidate. Their candidate logarithms range from approximately -9.98 to -0.80 , with median about -3.25 . The two failures occur in the left channel at $\sigma = 0.08$, final phase, for thresholds 10^{-8} and 10^{-6} ; both have $\gamma \approx 3.0719$. Thus the five-scale data do not support a claim that the local four-volume mechanism has disappeared.

The cross-scale conclusion is sharply different. Of 96 adjacent pairs, 67 give a positive transferred lower bound. Twenty-four have $b_* = +\infty$ because the target tail has greater rank than the source tail. Five additional pairs have finite b_* but $\sqrt{b_*}\gamma_{\text{source}} \geq 1$. Every one of the 24 phase/side/ threshold chains contains at least one such blocked edge, so the complete floor-free count is 0/24, not the regularized 24/24 reported in RH-125.

The obstruction is concentrated rather than random. Six infinite factors occur on the $0.16 \rightarrow 0.08$ edge, twelve on $0.08 \rightarrow 0.04$, and six on $0.02 \rightarrow 0.01$. The five finite superunit transfers all occur on $0.08 \rightarrow 0.04$. On the remaining $0.04 \rightarrow 0.02$ edge all 24 transfers are positive. This pattern is exactly what a changing finite-memory tail rank should produce; it is not explained by loss of recent-action rank.

5 Interpretation and revised route

RH-130 changes the status of the directional route in two ways. First, the positive Gram floor in RH-121 was not needed to preserve recent-action rank on the frozen data. The four-dimensional local packet remains visible. Second, the positive tail floor was mathematically consequential: it converted forbidden rank-creating transitions into finite, sometimes astronomical, multiplicative factors. The maximum factor near 10^{290} in RH-121 was therefore a warning sign, not merely an inconvenient condition number.

The correct next recurrence is not

$$D_{n+1} \preceq b_n S_n^* D_n S_n$$

but

$$D_{n+1} \preceq b_n S_n^* D_n S_n + F_n,$$

where $F_n \succeq 0$ covers tail directions born at the new scale. After normalization this is precisely the affine structure isolated in RH-123, $\gamma_{n+1}^2 \leq \rho_n \gamma_n^2 + q_n$. The immediate theoretical task is therefore to characterize the minimal forcing on the support complement, derive it from the memory recursion, and test whether q_n is small enough to coexist with eventual contraction.

This paper proves a finite-dimensional minimax theorem and a sharp rank-creation obstruction, and it invalidates the interpretation of the regularized five-scale chains as floor-stable evidence. It does not prove an all-level affine recurrence, a uniform bound on ρ_n or q_n , a positive normalized-base liminf, uniform Stage A, a Hilbert–Polya operator, an identification of zeta zeros, or the Riemann Hypothesis.

References

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